

Assignment 1

Exercise 1

Find velocity $v(t)$

- given is $a(t) = A - \kappa v(t)$
 - we know that $a(t) = \frac{dv}{dt}$, so
 - $\frac{dv}{dt} = A - \kappa v$
 - separate variables:
 - $\frac{dv}{A - \kappa v} = dt$
- integrate both sides:
 - use (u) -substitution: $u = A - \kappa v \Rightarrow du = -\kappa, dv$
 - $\int \frac{1}{A - \kappa v}, dv = \int dt$
 - left side $\Rightarrow -\frac{1}{\kappa} \ln |A - \kappa v|$, right side $\Rightarrow t + C_1$
 - so $-\frac{1}{\kappa} \ln |A - \kappa v| = t + C_1$
- now solve for $v(t)$:
 - $\ln |A - \kappa v| = -\kappa t - \kappa C_1$
 - $|A - \kappa v| = e^{-\kappa C_1} e^{-\kappa t}$
- absorb the sign and constant:
 - let $C_2 = (+-)e^{-\kappa C_1}$
 - $A - \kappa v = C_2 e^{-\kappa t}$
 - $\kappa v = A - C_2 e^{-\kappa t}$
 - $v(t) = \frac{A}{\kappa} - \frac{C_2}{\kappa} e^{-\kappa t}$
- so we finally have:

$$v(t) = \frac{A}{\kappa} - \frac{C_2}{\kappa} e^{-\kappa t}$$

- now we have to determine C_2
- we say that: $v(0) = v_0$:
 - $v_0 = v(0) = \frac{A}{\kappa} - \frac{C_2}{\kappa}$
 - $C_2 = A - \kappa v_0$
- substitute back:
 - $v(t) = \frac{A}{\kappa} - \frac{A - \kappa v_0}{\kappa} e^{-\kappa t}$

- rearrange:

$$v(t) = \frac{A}{\kappa} + \left(v_0 - \frac{A}{\kappa}\right)e^{-\kappa t}$$

Find position $x(t)$

- we know that: $\frac{dx}{dt} = v(t)$
- so we have $\frac{dx}{dt} = \frac{A}{\kappa} + \left(v_0 - \frac{A}{\kappa}\right)e^{-\kappa t}$
- we can integrate both sides to get $x(t)$:
 - $x(t) = \int \frac{A}{\kappa} dt + \int \left(v_0 - \frac{A}{\kappa}\right)e^{-\kappa t} dt$
- the first integral:
 - $\int \frac{A}{\kappa} dt = \frac{A}{\kappa}t$
- the second integral:
 - we know $\int e^{ct} dt = \frac{1}{c}e^{ct}$
 - so we get $\int \left(v_0 - \frac{A}{\kappa}\right)e^{-\kappa t} dt = \left(v_0 - \frac{A}{\kappa}\right)\left(-\frac{1}{\kappa}e^{-\kappa t}\right)$
- so finally we have:

$$x(t) = \frac{A}{\kappa}t - \frac{1}{\kappa}\left(v_0 - \frac{A}{\kappa}\right)e^{-\kappa t} + C_3$$

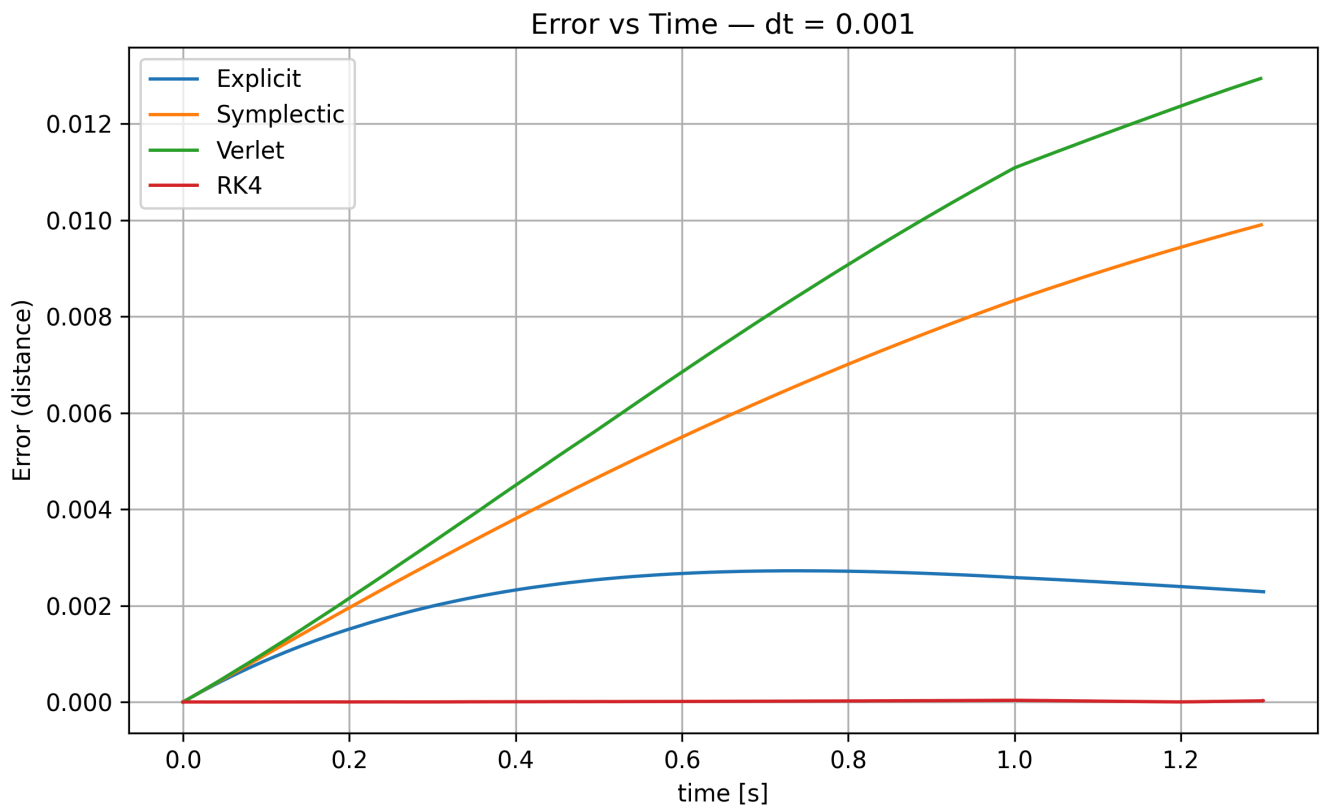
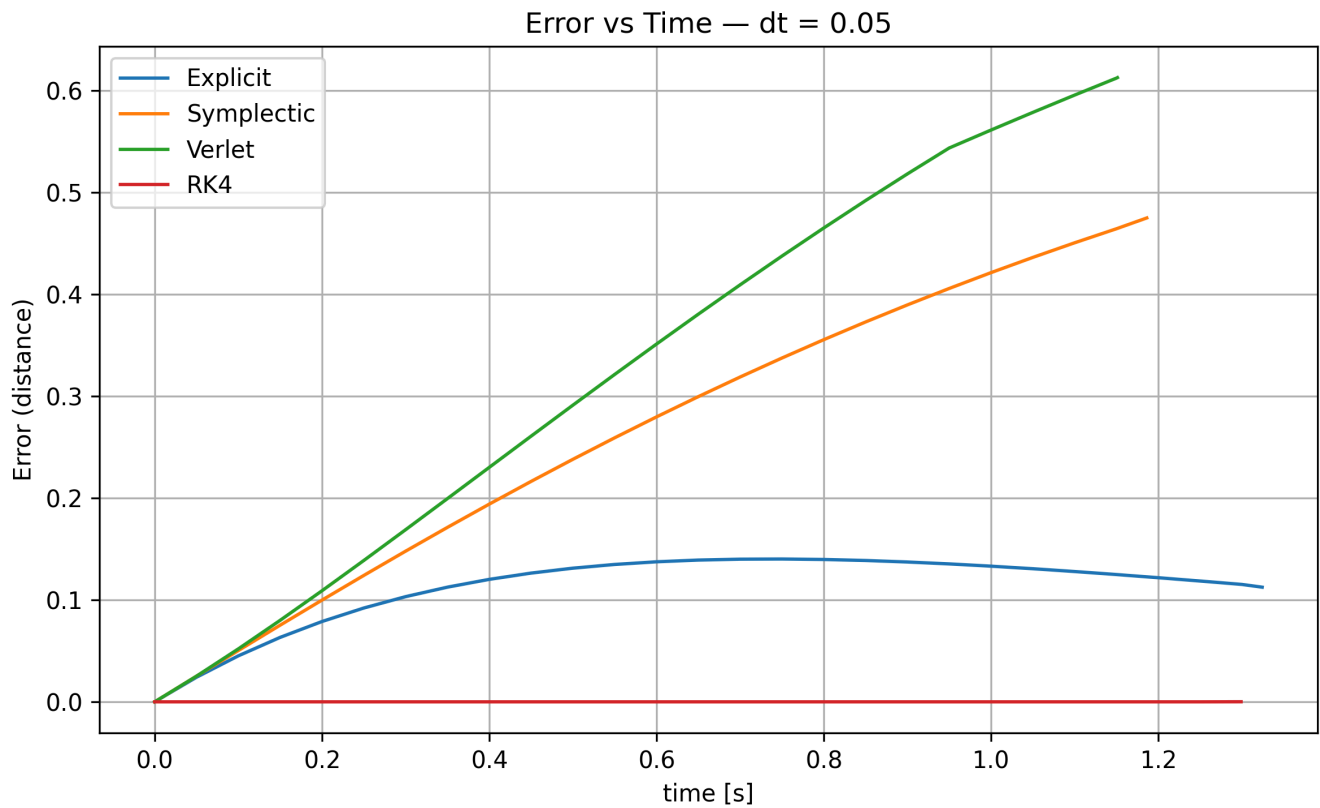
- no we need to find out C_3
- we define: $x(0) = x_0$
 - $x_0 = x(0) = -\frac{1}{\kappa}\left(v_0 - \frac{A}{\kappa}\right) + C_3$
 - $C_3 = x_0 + \frac{1}{\kappa}\left(v_0 - \frac{A}{\kappa}\right)$
- finally we substitute C_3 back:

$$x(t) = \frac{A}{\kappa}t - \frac{1}{\kappa}\left(v_0 - \frac{A}{\kappa}\right)e^{-\kappa t} + x_0 + \frac{1}{\kappa}\left(v_0 - \frac{A}{\kappa}\right)$$

- we can simplify this to:

$$x(t) = x_0 + \frac{1}{\kappa}\left(At + \left(v_0 - \frac{A}{\kappa}\right)(1 - e^{-\kappa t})\right)$$

Exercise 5

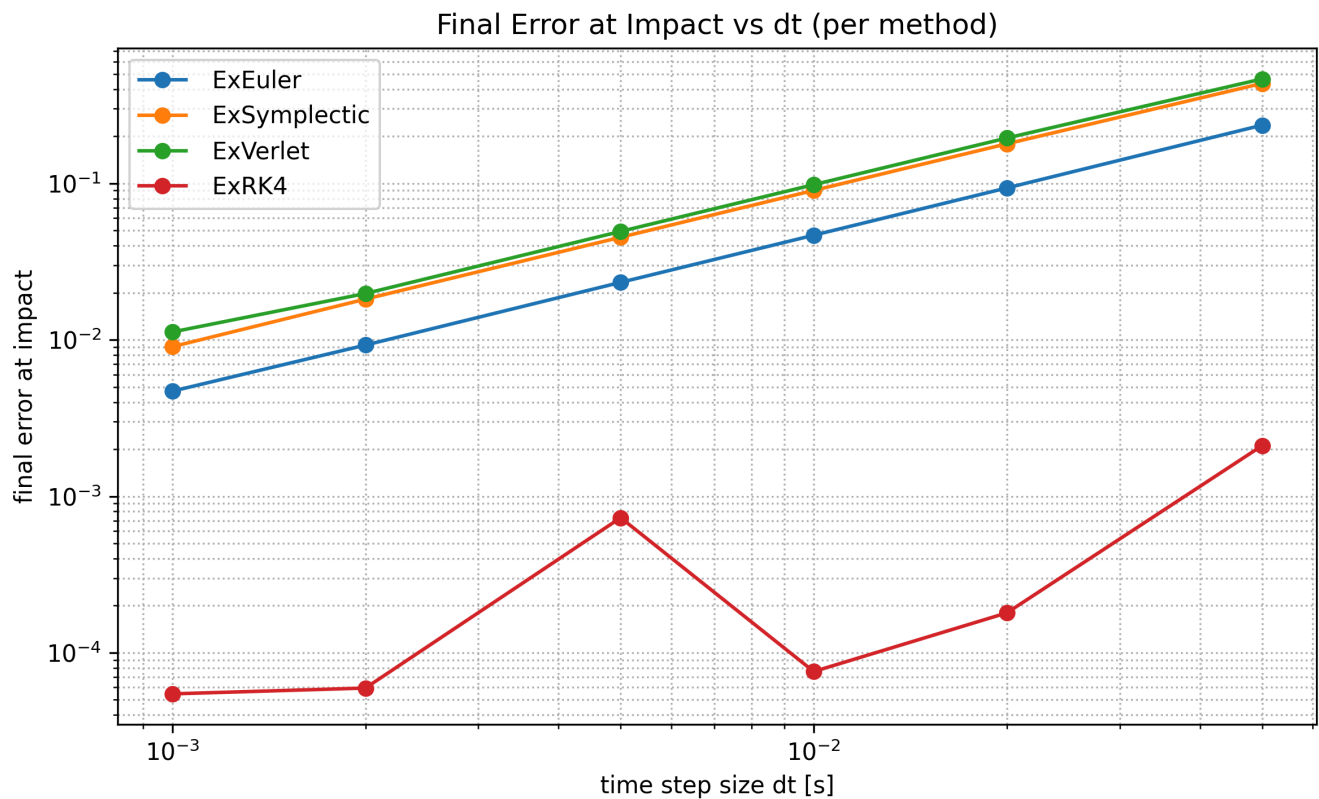


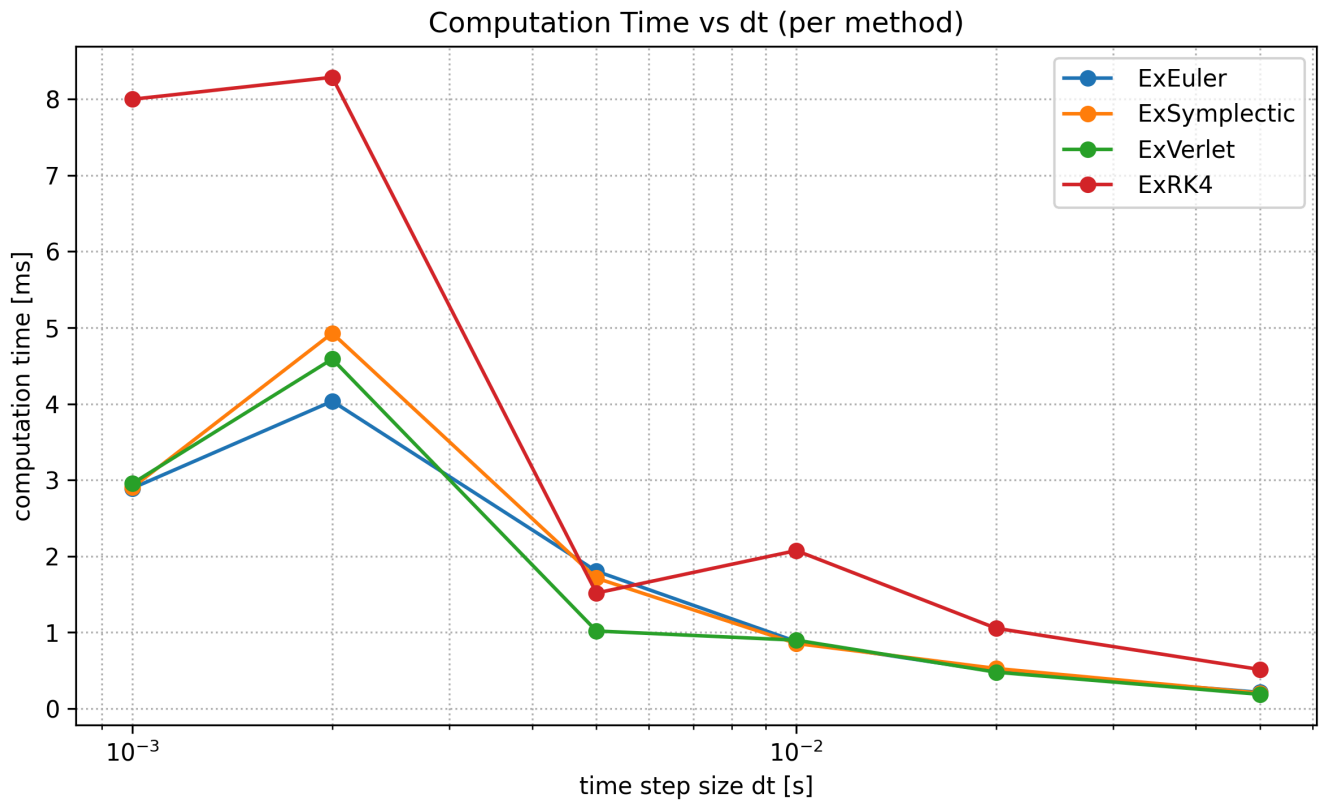
- we might have a mistake in the implementation of integration methods, as we expected Euler < Symplectic < Verlet < RK4
- but in our results Euler actually performs best after RK4

Exercise 6

- once damping increases, energy decreases
- all methods have sharp spikes when particles hit the ground
- energy smoothly decreases after each spike,
- with `damping=0.1` :
 - Euler: `max=95.3`
 - does not conserve energy
 - Symplectic: `max=11.9`
 - better at conserving energy
 - Verlet: `max=12.0`
 - better at conserving energy
 - RK4: `max=11.1`
 - best at conserving energy

Exercise 8





- for $dt = 10^{-2}$ RK4 seems to be the best choice regarding error and computation time
- before that (if computation time is more important than accuracy), Euler might be a better choice due to RK4's high computation time for those step sizes
- as already mentioned, we don't know why Euler performs better than Symplectic and Verlet in our implementation